

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

Time (years)	Total amount (dollars)
0	670.00
1	674.02
2	678.06

Sara opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Sara opened the account and the total amount d , in dollars, in the account. If Sara made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and d ?

Answer

- A. $d = 0.006(1 + 670)^t$
- B. $d = 670(1 + 0.006)^t$
- C. $d = (1 + 0.006)^t$
- D. $d = (1 + 670)^t$

Correct Answer: B

Rationale

Choice B is correct. It’s given that the relationship between t and d is exponential. The table shows that the value of d increases as the value of t increases. Therefore, the relationship between t and d can be represented by an increasing exponential equation of the form $d = a(1 + b)^t$, where a and b are positive constants. The table shows that when $t = 0, d = 670$. Substituting 0 for t and 670 for d in the equation $d = a(1 + b)^t$ yields $670 = a(1 + b)^0$, which is equivalent to $670 = a(1)$, or $670 = a$. Substituting 670 for a in the equation $d = a(1 + b)^t$ yields $d = 670(1 + b)^t$. The table also shows that when $t = 1, d = 674.02$. Substituting 1 for t and 674.02 for d in the equation $d = 670(1 + b)^t$ yields $674.02 = 670(1 + b)^1$, or $674.02 = 670(1 + b)$. Dividing both sides of this equation by 670 yields $1.006 = 1 + b$. Subtracting 1 from both sides of this equation yields $b = 0.006$. Substituting 0.006 for b in the equation $d = 670(1 + b)^t$ yields $d = 670(1 + 0.006)^t$. Therefore, of the choices, choice B best represents the relationship between t and d .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.